

RRH + SONTHILLU WEBSITES

PRODUCT REQUIREMENTS & V1 BLUEPRINT

Two distinct real-estate brands • One connected ecosystem • Shared CRM authority

Version 1.0 Draft Blueprint

14 August 2026

Scope: RRH (commercial, agriculture, open plots and related property) + Sonthillu (residential housing)

Purpose: establish a comprehensive product, UX, data, search, recommendation, integration, security, analytics, hosting and implementation baseline before development. Confirmed decisions are recorded; proposed items are explicitly identified; unresolved items are not guessed.

1. Document Control

Item	Value
Status	Draft for business + technical review
Websites	readhrealhomeproperties.com; sonthilluconstructions.com
Source of truth	RRH-CRMS for property/business lifecycle
Customer accounts	Separate RRH and Sonthillu accounts/databases
Website language	English
AI Search	Multilingual + mixed-language
Hosting	Existing Hostinger Business plan
Proposed public stack	Next.js + TypeScript + React
CRM context	Express + TypeScript + Prisma + MySQL + React/Vite
V1 principle	Industry-standard essentials first; extensible, not over-engineered

Status legend

- CONFIRMED — explicitly decided.
- PROPOSED — recommended approach.
- TBD — must be confirmed before implementation.
- FUTURE — intentionally deferred.

2. Executive Summary

RRH and Sonthillu are two distinct but connected real-estate companies. RRH handles commercial properties, agricultural land, open plots and related real-estate opportunities. Sonthillu focuses on residential housing such as apartments/flats, independent houses and villas. Both support resale.

The websites must not be clones. They will share product capabilities and backend integration patterns while maintaining separate visual identity, content tone, UX personality and customer accounts. Cross-branding is approved but should be subtle and contextual.

RRH-CRMS remains the authoritative source for property existence, verification, lifecycle and publication eligibility. A property may be associated with one or both companies; publication is independently controlled per brand.

Both websites provide Normal Search and AI Search. AI interprets natural language and multilingual/mixed-language input into a structured requirement model. A deterministic matching/ranking engine then retrieves and ranks actual CRM-approved inventory. A recommendation layer provides exact, related, nearby, slightly-above-budget and no-result recovery options.

V1 prioritizes practical production features. Advanced ML personalization, complex AI agents and large-scale search infrastructure are deferred until actual usage justifies them.

3. Business & Brand Model

3.1 RRH

- General real-estate company focused on commercial properties, agricultural lands and open plots.
- Rentals are not handled.
- Resale is supported; owners can submit property for RRH to market/sell.
- Current market: Telangana; planned future expansion to other states.
- Separate logo, branding, content and customer account system.

3.2 Sonthillu

- Residential/housing-oriented company.
- Handles own houses, villas, apartments/flats and related residential inventory.

- Resale is supported; owners can submit property for Sonthillu to market/sell.
- Separate logo, branding, content and customer account system.

3.3 Connected ecosystem

- Cross-branding is approved by the owner.
- Cross-brand references should appear in footer, About and relevant contextual discovery moments, not dominate the UI.
- If a visitor is looking for a category primarily handled by the other company, the site may provide a contextual link to that company.
- A property can be associated with both companies and published independently per company.

4. Product Goals & Non-Goals

Goals

- Launch two professional real-estate websites.
- Make discovery the central experience.
- Provide familiar industry-standard Normal Search.
- Provide AI Search as a differentiator.
- Provide property/project pages that convert browsing into enquiries.
- Support seller/resale submission into CRM.
- Provide shortlist, compare and multi-property enquiry.
- Make search recommendations a production feature.
- Provide separate website admin panels for analytics/configuration.
- Keep CRM as business/property source of truth.
- Build for future expansion without overbuilding V1.

Non-goals for V1

- Rental marketplace.
- Direct seller-to-customer contact.
- Public seller identity.
- Replicating CRM operational workflows in the website.
- Full ML recommendation stack from day one.
- Running large AI inference on Hostinger Business.
- Duplicating authoritative property records in both website databases.
- Overly complex personalization before enough behavioral data exists.

5. Personas & Actors

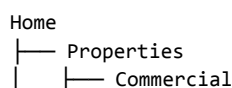
Actor	Goals	System
Anonymous visitor	Search, browse, compare, enquire	Public website
RRH customer	Save/search/enquire efficiently	RRH account
Sonthillu customer	Save/search/enquire efficiently	Sonthillu account
Seller/owner	Submit property for review	Website → CRM
Project Manager	Verify/enrich seller submission, site visit	CRM
Digital Marketing	Review photos/content/presentation	CRM
MD/approver	Final approval before publication	CRM
RRH website admin	Monitor search/AI/recommendations/content	RRH Admin
Sonthillu website admin	Monitor search/AI/recommendations/content	Sonthillu Admin
CRM/API	Authoritative business/property data	RRH-CRMS

6. Confirmed Business Rules

1. Rentals are not a website offering.
2. Sold and Reserved inventory cannot be live/public.
3. Inventory availability includes Available, Reserved and Sold.
4. Project lifecycle and inventory availability must be modeled separately.
5. New vs Resale is an explicit search dimension.
6. Source type is internal and never public by default.
7. The same physical property/project may be associated with RRH and Sonthillu.
8. Publication is independently controlled per company.
9. Exact property location is not public; approximate location is used.
10. CRM site verification can capture current latitude, longitude, accuracy, timestamp and employee identity.
11. Seller identity/contact details are never exposed to customers.
12. Prices are negotiable; do not present them as falsely fixed rates.
13. Price ranges may be displayed such as ₹45L – ₹53L.
14. RERA can be shown as a public verified indicator where applicable; detailed documents remain internal.
15. Logged-in customers can provide only preferred contact time when identity is already known.
16. Guests can login or use a guest enquiry form.
17. Both Request a Call and Call Now are available.
18. Users can enquire about multiple selected properties in one action.
19. Homepage primary discovery is Search + AI Search.
20. Sell Property is a main navbar action, not a dominant homepage hero action.
21. Website content/UI is English-only.
22. AI Search supports multilingual and mixed-language input.
23. RRH and Sonthillu have separate customer accounts/databases.
24. Core capabilities are shared, but UI/UX must be unique to each brand.
25. Stock imagery may be used during design where real assets are unavailable.

7. V1 Information Architecture

7.1 RRH



- Agricultural Land
- Open Plots
- Resale
- Projects
- Normal Search
- AI Search
- Sell Property
- Shortlist
- Compare
- About
- Contact
- Login / Account

7.2 Sonthillu

- Home
 - Properties
 - Apartments / Flats
 - Independent Houses
 - Villas
 - Resale
 - Projects
 - Normal Search
 - AI Search
 - Sell Property
 - Shortlist
 - Compare
 - About
 - Contact
 - Login / Account

Final category names must map to the actual CRM property taxonomy before implementation.

8. V1 Feature Matrix

Feature	RRH	Sonthillu	Priority	Authority/owner
Homepage	Yes	Yes	Must	Website
Normal Search	Yes	Yes	Must	Website + CRM data
AI Search	Yes	Yes	Must	AI + matching
Property results	Yes	Yes	Must	CRM/API
Property details	Yes	Yes	Must	CRM/API
Projects	Yes	Yes	Must/Recommended	CRM/API
Shortlist	Yes	Yes	Must	Website DB
Compare	Yes	Yes	Must	Website
Request a Call	Yes	Yes	Must	CRM
Call Now	Yes	Yes	Must	Website/telephony
Multi-property enquiry	Yes	Yes	Must	CRM
Sell Property	Yes	Yes	Must	Website → CRM
Login	Separate	Separate	Must	Website DB
Recommendations	Yes	Yes	Must	Recommendation layer
Search analytics	Yes	Yes	Must	Admin
AI analytics	Yes	Yes	Must	Admin
Cross-branding	Yes	Yes	Must	Website/admin
SEO foundation	Yes	Yes	Must	Website
Advanced ML personalization	Later	Later	Future	Recommendation layer
Rentals	No	No	Out	—

9. Page Requirements

9.1 Homepage

- Distinct brand header/navigation.
- Prominent Normal Search.
- Prominent AI Search.
- Brand-appropriate property categories.
- Featured/curated properties where enabled.
- Recently added properties.
- Popular categories/locations when enough data exists.
- Trust/verification messaging.
- Subtle cross-brand reference.
- Company/about section.
- Contact CTA.
- Footer with legal and cross-brand information.
- Mobile responsive.

9.2 Category/property discovery pages

- SEO-friendly category introduction.
- Search/filter entry.
- Results grid/list.
- Sort controls.
- Map option where implemented.
- Recommendations.
- No-result recovery.
- Responsive behavior.

9.3 Search results

- Query summary.
- Active filter chips.
- Result count.
- Sort.
- Property cards.
- Shortlist.
- Compare.
- Approximate location.
- Price range.
- New/Resale.
- Relevant verification badge.
- Match percentage/explanation where meaningful.
- Related/close matches.
- Mobile list/grid.

9.4 Property details

- Image gallery.
- Title.
- Type and New/Resale.
- Price range.
- Negotiability wording.
- Public-safe location.
- Applicable specifications.
- Amenities.

- RERA verified badge where applicable.
- Description.
- Project relationship.
- Approximate map.
- Request a Call.
- Call Now.
- Shortlist.
- Compare.
- Related properties.
- No seller identity/contact.

9.5 Project details

- Project overview.
- Location.
- Project status.
- Construction status.
- RERA indicator.
- Amenities.
- Gallery.
- Inventory summary.
- Available types.
- Price range.
- Description.
- SEO metadata.
- Enquiry CTA.

9.6 Sell Property

- Explain review/approval process.
- Choose New/Resale where applicable.
- Choose property category.
- Basic details.
- No photo requirement at initial submission.
- Seller contact information for company follow-up.
- Consent/privacy.
- Submit to CRM.
- Confirmation/reference.
- Never publish directly from website.

9.7 Shortlist / Compare

- Save/remove properties.
- Open details.
- Move to compare.
- Enquire about selected properties.
- Unavailable properties clearly labeled.
- Compare only meaningful/applicable attributes.
- N/A is distinct from missing.

9.8 Login/account

- Separate RRH/Sonthillu authentication.
- Secure sessions.
- Brand-specific saved activity.
- No silent account merge.

9.9 About/Contact

- Brand-specific story.
- Company contact details.
- Approved office/location details.
- Subtle connected-company explanation.
- No confusing blending of identities.

10. Normal Search

10.1 Baseline filters

- Location.
- Property type/category.
- Buy intent.
- New/Resale/Any.
- Minimum budget.
- Maximum budget.
- Property-type-specific attributes.
- Size/area.
- Amenities/features.
- Project filters where applicable.
- Optional radius/nearby location.
- Sort by relevance, newest, price low-high, price high-low and other justified options.

10.2 Budget semantics

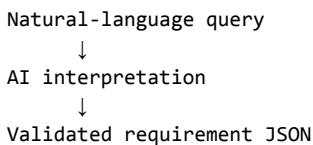
- ₹5L means ₹5 lakh purchase budget.
- If max budget is ₹50L, genuine matches at/below ₹50L should rank ahead of above-budget alternatives.
- Above-budget alternatives may appear only in a clearly labeled related section.
- Never silently change the user's maximum budget.
- Property ranges such as ₹45L–₹53L require overlap/fit rules.
- Negotiability is expected.

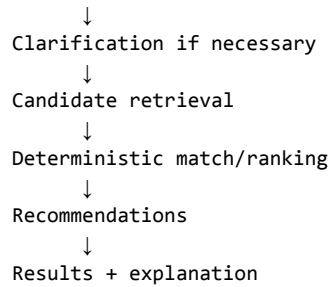
10.3 Applicability

Property	Useful attributes	Not inherently applicable
Apartment/Flat	BHK, floor, built-up/carpet area, parking, amenities	Agricultural acreage
Villa	BHK, built-up area, plot area, parking	Tower-specific fields
Independent House	BHK, built-up area, plot area, parking	Tower/unit fields
Open Plot	Plot area, dimensions, road width, facing, approvals	BHK
Agricultural Land	Acreage, land/access/water/road details	BHK
Commercial	Use type, area, frontage, floor, parking, approvals	BHK unless explicitly applicable

11. AI Search

11.1 Architecture principle





AI does not invent properties or make the final property truth decision.

11.2 Language

- English.
- Telugu.
- Hindi.
- Other supported languages where quality is acceptable.
- Mixed language.
- Telugu-English transliteration.
- Hindi-English mixing.
- Indian currency expressions: 5L, 50 lakhs, 0.5 crore, 1.2Cr, etc.
- Location synonyms and common locality expressions.

11.3 Example

User: 'Hyderabad lo 50 lakhs budget lo 2bhk kavalu'. The AI should identify Hyderabad, apartment/residential intent, 2BHK and ₹50L maximum, then pass normalized requirements to the matching engine.

11.4 Conceptual requirement model

```

{
  intent: BUY_PROPERTY,
  language: mixed,
  location: [...],
  propertyTypes: [...],
  listingType: ANY | NEW | RESALE,
  budget: { min, max, currency: INR },
  requirements: [...],
  preferences: [...],
  flexibleRequirements: [...],
  excludedRequirements: [...],
  unknownRequirements: [...],
  confidence: {...},
  clarificationNeeded: false
}
  
```

Final schema is a technical contract and must be versioned.

12. Dynamic Match & Ranking Engine

A fixed global weight table is rejected as the universal model. User requirements differ; property fields differ; some fields are not applicable.

12.1 Requirement classes

Class	Meaning	Behavior
Hard	Explicit must-have	Exclude incompatible candidates where verified
Strong preference	Important but flexible	High score contribution
Soft preference	Nice-to-have	Lower score contribution

Flexible	User allows variation	Alternative candidates permitted
Unknown	Not stated	No penalty solely due to absence
Not applicable	Does not apply	Ignore
Excluded	Explicitly unwanted	Exclude if reliably verifiable

12.2 Missing vs false vs N/A

Missing information is not automatically a negative. A plot has no BHK because BHK is not applicable. An apartment missing BHK is unknown. This distinction must be explicit in the engine.

12.3 Ranking

1. Determine actual requirements.
2. Identify hard constraints.
3. Retrieve viable candidates.
4. Evaluate each applicable requirement.
5. Ignore N/A fields.
6. Treat unknown data separately.
7. Aggregate satisfaction using only the requirements present.
8. Apply hard-constraint gating.
9. Apply public/availability/verification rules.
10. Rank exact matches, then close matches, then alternatives.

12.4 Result bands

- Exact/strong matches.
- Good matches with one flexible difference.
- Related matches.
- Slightly above budget / nearby alternatives.
- No-result recovery suggestions.

12.5 Tie-breakers

- Requirement satisfaction.
- Hard-constraint compliance.
- Verified/public-data confidence.
- User preferences.
- Freshness.
- Business-approved featured status only when it does not override user constraints.

13. Search & Property Recommendations

13.1 Types

Type	Example	Purpose
Search refinement	Try nearby locations	Recover weak/no-result searches
Related property	Similar unit at ₹52L	Continue discovery
Personalized	Based on recent activity	Return-visitor relevance

13.2 V1

- Similar properties.
- Close matches after search.
- Slightly above-budget alternatives.
- Nearby locations.
- Alternative property types when logically valid.
- Popular properties/categories for cold start.
- Recently viewed.

- Search-history suggestions where appropriate.
- No-result recovery.
- Search refinement suggestions.
- Recommendation impressions/clicks.

13.3 Behavioral signals

Signal	Indicative strength
Search	Low/medium
View	Medium
Shortlist	High
Compare	High
Enquiry	Very high
Site visit	Very high

These are signals, not absolute preferences. V1 may use explainable rules; ML is future.

13.4 Failed searches

Zero-result and low-result searches must be visible in admin analytics. They reveal unmet inventory demand and can inform sourcing, marketing and channel-partner activity.

13.5 Company boundaries

Recommendations on RRH must use only RRH-approved inventory; Sonthillu recommendations must use only Sonthillu-approved inventory. Cross-brand links are contextual, not silent inventory mixing.

14. Property & Project Data Requirements

14.1 Conceptual model

Property / Project

- Identity
- Company association/publication
- Classification
- Location
- Pricing
- Availability
- Project/construction status
- Specifications
- Amenities/features
- Media
- Verification
- Publication
- Search/SEO attributes

Field group	Examples	Public	Authority
Identity	ID, title, slug	Safe fields	CRM
Classification	Type, New/Resale	Yes	CRM
Company	RRH/Sonthillu publication	Controlled	CRM
Price	Min, max, INR, negotiable	Yes	CRM
Public location	State/city/locality/approx point	Yes	CRM
Exact location	Exact coordinates/address	No	CRM
Size	Built-up, carpet, plot, acreage	Yes	CRM
Specs	BHK, floors, facing, road width	When applicable	CRM
Amenities	Parking, lift, power backup	Yes	CRM
RERA	Number/status/verified badge	Safe subset	CRM
Media	Photos/videos	Yes	CRM/media
Availability	Available/Reserved/Sold	Available only live	CRM
Verification	Internal checks	Badge subset	CRM
Source type	Owner/channel partner/etc.	No	CRM

Publication	Draft/review/approved/published	No	CRM
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14.2 Status separation

Inventory: AVAILABLE | RESERVED | SOLD
 Project: UPCOMING | ACTIVE | COMPLETED | CLOSED
 Construction: UNDER_CONSTRUCTION | READY_TO_MOVE | N/A
 Publication: DRAFT | IN_REVIEW | APPROVED | PUBLISHED | UNPUBLISHED | REJECTED

Final enum names must match CRM rather than duplicate it.

Public live rule: Reserved and Sold inventory must not be exposed as live purchasable inventory.

15. Resale / Seller Workflow

```

Seller
↓
Website: Sell Property
↓
Register/login or guest seller intake
↓
New Property / Resale Property
↓
Basic details (no images required)
↓
CRM intake
↓
Project Manager assignment
↓
Site visit / verification
├─ amenities/details
├─ photos
├─ exact location
├─ current-location capture
└─ RERA/checks
↓
Digital Marketing review
├─ photos
└─ copy/spelling
↓
MD approval
↓
Company-specific publication
↓
Website live inventory
  
```

15.1 CRM location capture

- Occurs in CRM, not public website.
- 'Capture Current Location' control.
- Clear instruction to stand at property before capture.
- Permission prompt explanation.
- Store latitude, longitude, accuracy, timestamp, employee identity.
- Allow manual address confirmation/correction.
- Exact location remains internal.

16. CRM ↔ Website Integration

16.1 Authority

RRH Website → RRH Website API/BFF → RRH-CRMS

Sonthillu Website → Sonthillu Website API/BFF → RRH-CRMS

Customer accounts remain separate.

CRM remains authoritative for property/business lifecycle.

16.2 CRM → website

- Published properties by company.
- Projects.
- Inventory.
- Public attributes.
- Price ranges.
- Availability.
- Public location.
- RERA badge.
- Amenities.
- Media.
- SEO-safe content.
- Cross-brand configuration.

16.3 Website → CRM

- Enquiries.
- Request-a-call.
- Multi-property enquiry.
- Seller submission.
- Relevant customer references.
- Search/AI/recommendation analytics where approved.

16.4 Integration principles

- Version APIs.
- Typed contracts.
- Validate all input.
- Server-side company scoping.
- Rate limiting.
- Correlation IDs.
- Timeouts/retries only when safe.
- Caching for read-heavy data.
- Event/webhook invalidation later if needed.
- Never expose CRM internal fields.

17. Customer Database Architecture

RRH Web DB

- customers
- authentication/session
- shortlist
- compare
- website preferences/activity
- website-specific records

```

Sonthillu Web DB
├── customers
├── authentication/session
├── shortlist
├── compare
├── website preferences/activity
└── website-specific records

```

Do not duplicate authoritative property business records in these databases. A search index/cache is a derived read model, not a second source of truth.

18. Separate Website Admin Panels

18.1 RRH and Sonthillu

- Dashboard.
- Search analytics.
- AI Search analytics.
- Popular searches.
- Zero-result searches.
- Low-result searches.
- Popular locations.
- Property-type demand.
- Budget demand.
- Property views.
- Shortlists.
- Compare events.
- Enquiries.
- Recommendation performance.
- AI feature activation/configuration.
- Featured property/content controls where approved.
- Cross-brand links.
- Basic SEO/content controls.

18.2 Boundary

Admin panels monitor and configure the website; CRM continues to own verification, assignment, approval and operational property lifecycle.

19. Analytics & Event Tracking

Event	Minimum information
page_view	website, route, optional property/project ID
search_started	mode, session, time
normal_search	filters, result count
ai_search	query reference, language, normalized requirement version
property_view	property ID, source context
shortlist_add/remove	property ID
compare_add/remove	property ID
enquiry_start/submit	property IDs, contact preference
call_click	property/company context
sell_start/submit	category, submission reference
recommendation_impression	type, property ID
recommendation_click	type, property ID
refinement_click	suggestion type/value

- Do not store passwords in analytics.
- Do not expose seller PII.
- Use pseudonymous identifiers where practical.
- Define retention.
- Provide required consent/privacy mechanisms.

20. Proposed Technical Stack

Layer	Recommendation	Rationale
Public web	Next.js + TypeScript	SEO + React + SSR/SSG
UI	React + custom design system	Unique RRH/Sonthillu experiences
CSS	Tailwind CSS	Consistent implementation
Validation	Zod	Runtime/API contracts
Website DB	MySQL	Fits ecosystem
ORM	Prisma	Fits existing stack
BFF/API	Next.js server/API initially	Avoid unnecessary extra backend
CRM	Existing Express/TypeScript/Prisma/MySQL	Preserve source of truth
Search V1	MySQL + application matching + caching	Avoid premature search cluster
Matching	Custom TypeScript module/service	Deterministic and testable
Recommendations	Custom V1 rules/service	Explainable and extensible
AI	Provider/model abstraction	Avoid lock-in
Maps	Provider abstraction	Future flexibility
Hosting	Hostinger Business	Existing environment
Deployment	GitHub → Hostinger	Supported deployment path

20.1 Why Next.js

The CRM is an application; the public sites are search/discovery products with indexable property/project pages. Next.js is therefore proposed for SEO-friendly public pages while retaining React/TypeScript.

20.2 Why not WordPress

The product requires typed property APIs, dynamic search, AI interpretation, matching, recommendations, separate customer accounts, CRM integration and analytics. A custom TypeScript application is a cleaner fit.

20.3 AI hosting

Do not run a large AI model on the Hostinger Business web server. The AI inference layer should be independently deployable and accessed through a stable internal interface.

21. Hostinger Architecture

Hostinger's current documentation states that Business hosting supports Node.js web applications and lists Next.js, React and Vite among supported technologies. Its current Business listing shows 5 web apps, 2 CPU cores, 3 GB RAM, 50 GB NVMe, unlimited bandwidth, managed MySQL, GitHub integration, SSL, CDN, WAF and backups. These values must be rechecked in hPanel before deployment because plan entitlements can change.

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App	Role	Proposed
1	RRH website + admin	Hostinger Node.js app
2	Sonthillu website + admin	Hostinger Node.js app
3	Existing CRM	Existing deployment
4	Customer Portal	Existing deployment
5	Reserve/future	Avoid consuming without need

Hostinger documents GitHub deployment/automatic build flows for Node.js web apps and explicitly lists Next.js/Express support. 穀cite學turn0search0傢

21.1 Scaling

- V1 should target current plan.
- Monitor resource usage.
- Do not host AI inference on the shared web server.
- If traffic/search/database load grows, move compute/search to stronger infrastructure without changing product contracts.

22. Security & Privacy

- HTTPS everywhere.
- Secure HTTP-only sessions/cookies where used.
- CSRF protection where required.
- Strong password hashing if passwords are stored.
- Rate limiting.
- Runtime validation.
- Secure headers.
- Least-privilege API credentials.
- No sensitive tokens in browser localStorage.
- No seller details in public API.
- No exact coordinates in public payloads.
- Separate RRH/Sonthillu auth scopes.
- Audit sensitive admin actions.
- No internal source type/employee data publicly.
- Server-side authorization against IDOR.
- Secrets only in environment/secret management.
- Dependency scanning.
- Backups/recovery.

23. SEO

- Unique title/meta per property/project/category.
- Canonical URLs.
- Clean slugs.
- Open Graph metadata.
- XML sitemap.
- Robots.txt.
- Accurate structured data where appropriate.
- Index only publishable public pages.
- Noindex admin/account/internal filter states.
- Avoid filter-URL index explosion.
- Useful location/category content.
- Clear sold/unpublished redirect/404 policy.
- Performance/Core Web Vitals monitoring.

24. UI/UX

24.1 Shared

- Familiar real-estate patterns without copying competitor UI.
- Original layouts and components.
- Fast discovery.

- Mobile-first responsiveness.
- Accessible controls.
- Clear filters.
- Transparent recommendation labels.
- No silent modification of requirements.

24.2 RRH direction

- Professional/investment/property-business personality.
- Use logo-derived palette.
- Strong commercial/land/plot imagery.
- Modern, credible, business-oriented.

24.3 Sonthillu direction

- Residential/home/lifestyle personality.
- Use logo-derived palette.
- Lifestyle, home, villa and apartment imagery.
- Warm/premium/trust-oriented.

The final UI design must not be a color-swapped clone. Both sites should share quality and core capability, not visual identity.

25. Accessibility

- Semantic HTML.
- Keyboard navigation.
- Visible focus.
- Accessible labels/errors.
- Good contrast.
- Meaningful alt text.
- Decorative-image handling.
- Touch-friendly targets.
- Do not rely on color alone.
- Accessible dialogs/drawers.

26. Performance

- Optimize images and formats.
- Responsive image delivery.
- Lazy-load non-critical media.
- Minimize client JS.
- Cache public reads safely.
- Avoid repeated CRM calls.
- Pagination/incremental loading.
- Monitor Core Web Vitals.
- Separate AI latency from normal search UX.
- Skeleton/loading states.

27. Edge Cases & Failure States

Scenario	Required behavior
No results	Explain no exact match; show alternatives/refinements

Only above-budget matches	Separate and label them
Property reserved	Remove from live search after state propagation
Property sold	Remove from live search; follow SEO policy
Unpublished	Never expose through public API
BHK missing on land	N/A, no penalty
BHK missing on apartment	Unknown; handle according to hard/soft requirement
Exact location unavailable	Use safe locality/approximate location
CRM location permission denied	Manual correction path in CRM
AI ambiguity	Ask clarification or offer Normal Search
AI invalid output	Schema validation + retry/fallback
CRM unavailable	Controlled error; no internal leakage
Duplicate enquiry	Idempotency/deduplication
Unavailable shortlisted property	Keep historical record but label unavailable; suggest alternatives
Compare N/A field	Show N/A, not false
Image failure	Fallback image
Cross-brand unpublished property	Never leak through recommendation

28. Content & Media

- Brand-specific copy.
- CRM-approved property/project descriptions.
- Category content.
- About/contact.
- Privacy/terms/cookie notices as required.
- Accurate RERA wording.
- Real property photos in production preferred.
- Stock images only where they do not misrepresent a specific property.
- Responsive media variants.
- Primary image ordering.
- Safe alt text.
- Internal documents never public.

29. AI Production Requirements

- Normal Search always remains available.
- AI is optional.
- AI output must be schema validated.
- AI never invents properties, prices, availability or amenities.
- AI interprets requirements; deterministic engine decides property matching.
- Clarify ambiguous inputs.
- Provider/model abstraction.
- Model version tracking where appropriate.
- Design for model replacement.
- Avoid dependence on a single paid API when effectively unlimited usage is a business requirement.
- AI inference infrastructure is independent from Hostinger web hosting.

30. Candidate API Contract Inventory

Capability	Direction	Purpose
Published property search	CRM → website	Brand-scoped live inventory
Property detail	CRM → website	Public property data
Projects	CRM → website	Published projects
Project detail	CRM → website	Public project data
Property types	CRM → website	Canonical taxonomy
Amenities	CRM → website	Applicable filters

Locations	CRM → website	Search/location data
Enquiry	Website → CRM	Create lead/enquiry
Multi-property enquiry	Website → CRM	One enquiry for selections
Call request	Website → CRM	Follow-up
Seller submission	Website → CRM	Initial resale/new submission
Publication event	CRM → website	Cache invalidation/sync
Search event	Website → analytics	Demand tracking
AI event	Website → analytics	AI quality/adoption
Recommendation event	Website → analytics	Recommendation performance

These are requirement-level operations, not final endpoint names. Existing CRM APIs must be inspected before implementation.

31. Testing Strategy

Functional

- Normal Search.
- AI Search.
- Property results/details.
- Projects.
- Shortlist.
- Compare.
- Login.
- Guest enquiry.
- Logged-in enquiry.
- Multi-property enquiry.
- Sell Property.
- Cross-brand links.
- Admin analytics.

Search/matching

- Budget boundaries.
- Price-range overlap.
- New/Resale.
- Location synonyms.
- Property-type applicability.
- Missing vs N/A.
- Hard/soft/flexible requirements.
- No results.
- Above-budget alternatives.
- Ranking consistency.
- Recommendations.

AI

- English.
- Telugu.
- Hindi.
- Mixed-language.
- Transliteration.
- Currency expressions.
- Ambiguous location.
- Ambiguous type.

- Missing requirements.
- Contradictory requirements.
- Invalid model output.
- Prompt injection/adversarial inputs.
- Hallucination prevention.
- Fallback.

Integration

- CRM timeout/unavailable.
- Stale property state.
- Reserved/Sold transition.
- API authorization.
- Duplicate submission.
- Invalid payload.
- Cross-brand isolation.

32. Development & Deployment

- Separate applications/repos or clearly isolated projects for RRH and Sonthillu.
- Development → staging/UAT → production where feasible.
- Environment variables for secrets/config.
- No secrets in source control.
- GitHub-based deployment.
- Intentional DB migrations.
- Health checks.
- Redacted logs.
- Backups.
- Rollback plan.
- Production monitoring.

```

Development
  ↓
Staging / UAT
  ↓
Production

RRH Web + Admin
Sonthillu Web + Admin
RRH-CRMS
Customer Portal

```

33. V1 Delivery Phases

Phase	Scope	Exit
P0 Discovery	Inspect CRM APIs, lifecycle, schema, Hostinger, domains	Integration constraints confirmed
P1 Design	Two brand design systems + responsive shell	Approved UI direction
P2 Foundation	Home, nav, footer, routing, SEO foundation	Public shell complete
P3 Discovery	Normal Search, filters, results, details, projects	Normal search end-to-end
P4 Customer	Login, shortlist, compare, enquiry/call	Conversion flow complete
P5 Seller	Sell Property → CRM	Submission safe and traceable
P6 Recommendations	Similar/related/no-result + analytics	Measured recommendations
P7 AI	AI interpretation + multilingual + matching	Safe AI Search
P8 Admin	Analytics/configuration	Admin visibility

P9 Hardening	Security/performance/accessibility/SEO/edge cases	Production-ready
P10 Launch	Deploy, monitor, backup, rollback	Live

34. Acceptance Criteria

1. RRH and Sonthillu are visually and experientially distinct.
2. Separate domains and customer databases work correctly.
3. CRM remains property/business source of truth.
4. Only brand-approved live inventory is public.
5. Reserved/Sold inventory is not live.
6. Normal Search works without AI.
7. AI Search supports multilingual/mixed input.
8. AI output is validated and cannot invent property data.
9. Ranking is requirement-aware, not a fixed global weight table.
10. N/A fields do not cause false mismatch penalties.
11. Price ranges and negotiability are represented correctly.
12. Exact location is protected.
13. RERA public badge is safe and accurate.
14. Seller details are private.
15. Shortlist and Compare work.
16. Multi-property enquiry works.
17. Guest and logged-in enquiry flows work.
18. Seller submissions reach CRM and do not publish automatically.
19. Recommendations cover exact/related/no-result scenarios.
20. Analytics capture search/recommendation outcomes.
21. Admin is separate by brand and does not duplicate CRM operations.
22. SEO basics are implemented.
23. Mobile UX is production quality.
24. Cross-brand data leakage is prevented.
25. Deployment fits the approved hosting architecture.

35. Open Questions / Pre-Implementation Decisions

1. Exact final property taxonomy in CRM for RRH and Sonthillu.
2. Existing CRM project/unit model and reusable APIs.
3. Existing CRM public property APIs.
4. Existing CRM lifecycle/publication enum names.
5. Whether a property can have different public titles/descriptions per brand.
6. Exact phone/WhatsApp Call Now policy.
7. Final guest enquiry required fields and consent wording.
8. Whether site-visit booking is V1.
9. Map provider.
10. WhatsApp contact/enquiry in V1 or later.

11. Blog/content CMS requirement.
12. Final legal/privacy/cookie wording.
13. Analytics provider vs first-party-only.
14. AI model/infrastructure selection after evaluation.
15. Recommendation data retention.
16. Login method: OTP/password/etc.
17. Separate MySQL databases vs database servers for website accounts.
18. Current Hostinger web-app slot usage.
19. Staging hosting arrangement.
20. Final transparent/SVG logo assets.

36. Risks & Mitigations

Risk	Impact	Mitigation
CRM API mismatch	High	Inspect CRM before coding; version contracts
Taxonomy mismatch	High	Canonical CRM-to-website mapping
AI hallucination	High	Structured AI output + deterministic matcher
AI cost/limits	High	Model abstraction + controlled/self-hosted evaluation
Hostinger limits	Medium/High	Monitor; keep AI off web server; scale if needed
Stale availability	High	Short cache + invalidation/events
Cross-brand leakage	Critical	Server-side brand authorization
Poor recommendations	Medium	Explainable V1 + measurement + iteration
SEO duplicates	Medium	Canonical/noindex/index strategy
Image bloat	Medium	Optimization/CDN
Scope creep	High	Strict V1/Future classification

37. Change Control

- Every new feature is classified as V1 Must, V1 Recommended, V1.1 or Future.
- Do not guess business rules affecting visibility, pricing, legal status or privacy.
- CRM ownership/publication changes require API/data-contract review.
- Customer identity changes require security review.
- AI changes must preserve structured requirement boundary.
- UI changes must preserve brand separation and accessibility.

38. Glossary

Term	Definition
CRM	RRH-CRMS, authoritative operational/business system
Normal Search	Structured real-estate filter search
AI Search	Natural-language interpretation layer
Match Engine	Deterministic property requirement scoring/ranking
Recommendation Engine	System for related properties/refinements/personalization
Publishable	Approved and authorized for a specific brand
Available	Inventory currently public as available

Resale	Third-party-owned property submitted for marketing/sale
Source type	Internal property origin
Approximate location	Public-safe location without exact property coordinates
Hard requirement	Must-have condition
Unknown	Data unavailable; not false
N/A	Field does not apply to this property type

39. External Hosting Reference

Hostinger documentation was consulted for the current hosting assumptions. Hostinger states that Business hosting supports Node.js web applications and lists Next.js, React and Vite among supported technologies. The current Business plan page lists 5 web apps, 2 CPU cores, 3 GB RAM, 50 GB NVMe, unlimited bandwidth, managed MySQL, GitHub integration, SSL, CDN, WAF and backups. These values must be verified in the user's actual hPanel before deployment because commercial plan details can change. 穀cite學turn0search0學turn0search1係

References: 穀url學Hostinger Node.js Web App documentation學turn0search0係 and 穀url學Hostinger Business Web Apps Hosting學turn0search1係

40. Final Product Principle

The websites should be simple for customers and sophisticated behind the scenes. The user should experience familiar real-estate discovery, clear property information, fast Normal Search, useful recommendations and an optional AI Search. Complexity should live in the CRM integration, data contracts, matching engine, recommendation layer and analytics rather than unnecessary UI.

RRH and Sonthillu are separate brands. They should look and feel different while providing the same level of product quality. Their shared foundation is the connected CRM/API ecosystem, not a shared visual identity.

This blueprint is the product baseline. The next artifact after review should reconcile it against the actual RRH-CRMS repository, database schema, lifecycle and existing API surface, then produce the final implementation-ready requirements/data-contract document.